

Browser tabs: Your Repositories, Untitled5.ipynb - Colab, Cloud Shell

Address bar: shell.cloud.google.com/?show=ide%2Cterminal

Cloud Shell Editor

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 Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#)

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```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
dgmidhun@cloudshell:~$ uname
Linux
dgmidhun@cloudshell:~$ uptime
 06:29:32 up 33 min,  0 user,  load average: 2.08, 0.67, 0.26
dgmidhun@cloudshell:~$ hostname
cs-892642682939-default
dgmidhun@cloudshell:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
dgmidhun@cloudshell:~$
```

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PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2446	dgmidhun	20	0	1905780	666908	47324	R	64.0	8.2	0:07.66	node
2430	dgmidhun	20	0	11.6g	459056	54252	R	62.0	5.6	0:07.61	node
1981	dgmidhun	20	0	42.7g	911400	62132	R	61.7	11.2	0:24.96	node
734	dgmidhun	20	0	11.3g	158424	51252	S	0.3	1.9	0:06.09	node
1504	dgmidhun	20	0	1201424	76680	48212	S	0.3	0.9	0:01.14	node
1	root	20	0	4324	3364	3084	S	0.0	0.0	0:00.06	bash
8	syslog	20	0	222140	3852	2496	S	0.0	0.0	0:00.01	rsyslogd
24	root	20	0	38768	28052	10948	S	0.0	0.3	0:00.53	python
25	root	20	0	6196	2680	2440	S	0.0	0.0	0:00.00	logger
66	root	10	-10	12024	4188	3076	S	0.0	0.1	0:00.00	sshd
318	root	20	0	2021836	84528	61272	S	0.0	1.0	0:00.40	dockerd
413	root	20	0	1234268	13920	8340	S	0.0	0.2	0:00.30	editor-proxy
414	root	20	0	16328	6452	5532	S	0.0	0.1	0:00.02	sudo
425	root	20	0	1226100	2728	1976	S	0.0	0.0	0:00.01	tmux-agent
453	root	20	0	1793224	46276	32572	S	0.0	0.6	0:00.51	containerd
647	root	20	0	2696	1584	1484	S	0.0	0.0	0:00.00	sleep
674	dgmidhun	10	-10	7148	2868	2220	S	0.0	0.0	0:00.01	tmux
675	dgmidhun	10	-10	8800	6020	4152	S	0.0	0.1	0:00.07	bash
699	root	20	0	9196	4004	3624	S	0.0	0.0	0:00.00	runuser
700	dgmidhun	20	0	4324	3332	3064	S	0.0	0.0	0:00.01	sh
1515	dgmidhun	20	0	9024	6000	3980	S	0.0	0.1	0:00.06	bash
1966	root	10	-10	13996	9664	8288	S	0.0	0.1	0:00.02	sshd
1968	root	10	-10	13996	9748	8372	S	0.0	0.1	0:00.01	sshd
1970	dgmidhun	10	-10	14256	6500	4844	S	0.0	0.1	0:00.00	sshd
1971	dgmidhun	10	-10	14256	6368	4712	S	0.0	0.1	0:00.00	sshd
1972	dgmidhun	10	-10	7340	3732	3484	S	0.0	0.0	0:00.00	bash
1975	dgmidhun	10	-10	7604	4572	3956	S	0.0	0.1	0:00.00	bash
1978	dgmidhun	10	-10	7340	3676	3424	S	0.0	0.0	0:00.00	bash
1979	dgmidhun	10	-10	7340	3684	3420	S	0.0	0.0	0:00.00	start-shell.sh
1980	dgmidhun	10	-10	6880	3340	3040	S	0.0	0.0	0:00.00	tmux
1992	dgmidhun	20	0	1048516	61708	47896	S	0.0	0.8	0:00.24	node
2015	dgmidhun	20	0	1016392	63032	46372	S	0.0	0.8	0:00.24	node
2024	dgmidhun	20	0	9024	5984	3956	S	0.0	0.1	0:00.07	bash
2472	dgmidhun	10	-10	11932	5684	3468	R	0.0	0.1	0:00.00	top

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```
1.3%] Tasks: 42, 103 thr, 0 kthr; 1 running
1.3%] Load average: 1.33 0.63 0.26
2.52G/7.76G] Uptime: 00:34:15
OK/OK]
```

F1 Help F2 Setup F3 Search F4 Filter F5 Free F6 Sort By F7 Nice - F8 Nice + F9 Kill F10 Quit

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```
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You can view your projects by running `gcloud projects list`.
dgmidhun@cloudshell:~$ lsb_release
No LSB modules are available.
dgmidhun@cloudshell:~$ arch
x86_64
dgmidhun@cloudshell:~$ uname
Linux
dgmidhun@cloudshell:~$ ls
README-cloudshell.txt
dgmidhun@cloudshell:~$ cd
dgmidhun@cloudshell:~$ pwd
/home/dgmidhun
dgmidhun@cloudshell:~$ mkdir
mkdir: missing operand
Try 'mkdir --help' for more information.
dgmidhun@cloudshell:~$ rmdir
rmdir: missing operand
Try 'rmdir --help' for more information.
dgmidhun@cloudshell:~$ touch
touch: missing file operand
Try 'touch --help' for more information.
dgmidhun@cloudshell:~$ rm
rm: missing operand
Try 'rm --help' for more information.
dgmidhun@cloudshell:~$ cp
cp: missing file operand
Try 'cp --help' for more information.
dgmidhun@cloudshell:~$ mv
mv: missing file operand
Try 'mv --help' for more information.
dgmidhun@cloudshell:~$ ln
ln: missing file operand
Try 'ln --help' for more information.
dgmidhun@cloudshell:~$
```

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```
dgmidhun@cloudshell:~$ sed
Usage: sed [OPTION]... {script-only-if-no-other-script} [input-file]...

-n, --quiet, --silent
    suppress automatic printing of pattern space
--debug
    annotate program execution
-e script, --expression=script
    add the script to the commands to be executed
-f script-file, --file=script-file
    add the contents of script-file to the commands to be executed
--follow-symlinks
    follow symlinks when processing in place
-i[SUFFIX], --in-place[=SUFFIX]
    edit files in place (makes backup if SUFFIX supplied)
-l N, --line-length=N
    specify the desired line-wrap length for the 'l' command
--posix
    disable all GNU extensions.
-E, -r, --regexp-extended
    use extended regular expressions in the script
    (for portability use POSIX -E).
-s, --separate
    consider files as separate rather than as a single,
    continuous long stream.
--sandbox
    operate in sandbox mode (disable e/r/w commands).
-u, --unbuffered
    load minimal amounts of data from the input files and flush
    the output buffers more often
-z, --null-data
    separate lines by NUL characters
--help
    display this help and exit
--version
    output version information and exit
```

If no `-e`, `--expression`, `-f`, or `--file` option is given, then the first non-option argument is taken as the sed script to interpret. All remaining arguments are names of input files; if no input files are specified, then the standard input is read.

GNU sed home page: <<https://www.gnu.org/software/sed/>>.
General help using GNU software: <<https://www.gnu.org/gethelp/>>.
dgmidhun@cloudshell:~\$

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```
load minimal amounts of data from the input files and flush
the output buffers more often
-z, --null-data
    separate lines by NUL characters
--help
    display this help and exit
--version
    output version information and exit

If no -e, --expression, -f, or --file option is given, then the first
non-option argument is taken as the sed script to interpret. All
remaining arguments are names of input files; if no input files are
specified, then the standard input is read.

GNU sed home page: <https://www.gnu.org/software/sed/>.
General help using GNU software: <https://www.gnu.org/gethelp/>.
dgmidhun@cloudshell:~$ chmod
chmod: missing operand
Try 'chmod --help' for more information.
dgmidhun@cloudshell:~$ chown
chown: missing operand
Try 'chown --help' for more information.
dgmidhun@cloudshell:~$ ghgrp
-bash: ghgrp: command not found
dgmidhun@cloudshell:~$ chgrp
chgrp: missing operand
Try 'chgrp --help' for more information.
dgmidhun@cloudshell:~$ umask
0002
dgmidhun@cloudshell:~$ stst
-bash: stst: command not found
dgmidhun@cloudshell:~$ stat
stat: missing operand
Try 'stat --help' for more information.
dgmidhun@cloudshell:~$ getfacl
-bash: getfacl: command not found
dgmidhun@cloudshell:~$ setfacl
-bash: setfacl: command not found
dgmidhun@cloudshell:~$ lsattr
-----e----- ./README-cloudshell.txt
dgmidhun@cloudshell:~$ chattr
Usage: chattr [-RVf] [-+=aAcCdDeijPsStTuFvX] [-p project] [-v version] files...
dgmidhun@cloudshell:~$ usermod
Usage: usermod [options] LOGIN
```

Gemini CLI is available in Cloud Shell terminal! Type `gemini` to try it. [Learn more](#) Don't show again Dismiss

```
dgmidhun@cloudshell:~$ setfacl
-bash: setfacl: command not found
dgmidhun@cloudshell:~$ lsattr
-----e----- ./README-cloudshell.txt
dgmidhun@cloudshell:~$ chattr
Usage: chattr [-RVf] [--aAcCdDeijPsStTuFh] [-p project] [-v version] files...
dgmidhun@cloudshell:~$ usermod
Usage: usermod [options] LOGIN

Options:
-a, --append                append the user to the supplemental GROUPS
                             mentioned by the -G option without removing
                             the user from other groups
-b, --badname               allow bad names
-c, --comment COMMENT      new value of the GECOS field
-d, --home HOME_DIR        new home directory for the user account
-e, --expiredate EXPIRE_DATE set account expiration date to EXPIRE_DATE
-f, --inactive INACTIVE    set password inactive after expiration
                             to INACTIVE
-g, --gid GROUP             force use GROUP as new primary group
-G, --groups GROUPS        new list of supplementary GROUPS
-h, --help                  display this help message and exit
-l, --login NEW_LOGIN      new value of the login name
-L, --lock                  lock the user account
-m, --move-home             move contents of the home directory to the
                             new location (use only with -d)
-o, --non-unique            allow using duplicate (non-unique) UID
-p, --password PASSWORD    use encrypted password for the new password
-P, --prefix PREFIX_DIR    prefix directory where are located the /etc/* files
-r, --remove                remove the user from only the supplemental GROUPS
                             mentioned by the -G option without removing
                             the user from other groups
-R, --root CHROOT_DIR      directory to chroot into
-s, --shell SHELL          new login shell for the user account
-u, --uid UID               new UID for the user account
-U, --unlock                unlock the user account
-v, --add-subuids FIRST-LAST add range of subordinate uids
-V, --del-subuids FIRST-LAST remove range of subordinate uids
-w, --add-subgids FIRST-LAST add range of subordinate gids
-W, --del-subgids FIRST-LAST remove range of subordinate gids
-Z, --selinux-user SEUSER  new SELinux user mapping for the user account

dgmidhun@cloudshell:~$
```

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```
dgmidhun@cloudshell:~$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1460
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 3e:6c:14:c3:5f:aa txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.88.0.3 netmask 255.255.0.0 broadcast 10.88.255.255
    ether 9e:46:5e:36:19:e1 txqueuelen 1000 (Ethernet)
    RX packets 13555 bytes 39902877 (39.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 9831 bytes 40697400 (40.6 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 10808 bytes 88032574 (88.0 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 10808 bytes 88032574 (88.0 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

dgmidhun@cloudshell:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where OBJECT := { address | addrlabel | amt | fou | help | ila | ioam | l2tp |
                  link | macsec | maddress | monitor | mptcp | mroute | mrule |
                  neighbor | neighbour | netconf | netns | nexthop | ntable |
                  ntbl | route | rule | sr | tap | tcpmetrics |
                  token | tunnel | tuntap | vrf | xfrm }
OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
             -h[uman-readable] | -iec | -j[son] | -p[retty] |
             -f[amily] { inet | inet6 | mpls | bridge | link } |
             -4 | -6 | -M | -B | -O |
             -l[oops] { maximum-addr-flush-attempts } | -br[ief] |
             -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
             -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
             -c[olor] }
```

```

dgmidhun@cloudshell:~$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address          State
tcp        0      0 cs-892642682939-d:40690 sm-in-fl39.1e100.:https ESTABLISHED
tcp        0      0 localhost:47354          localhost:971            ESTABLISHED
tcp        0    180 localhost:ssh            localhost:53420          ESTABLISHED
tcp        0      0 localhost:47364          localhost:971            ESTABLISHED
tcp        0      0 localhost:971            localhost:47354          ESTABLISHED
tcp        0      0 localhost:55410          localhost:970            ESTABLISHED
tcp        0      0 localhost:971            localhost:47364          ESTABLISHED
tcp        0      0 localhost:37950          localhost:ssh            TIME_WAIT
tcp        0      0 cs-892642682939-d:51252 cdn-185-199-110-15:http ESTABLISHED
tcp        0      0 localhost:53420          localhost:ssh            ESTABLISHED
tcp        0      0 localhost:55396          localhost:970            ESTABLISHED
tcp        0      0 localhost:ssh            localhost:55138          ESTABLISHED
tcp        0      0 localhost:55138          localhost:ssh            ESTABLISHED
tcp6       0      0 cs-892642682939-def:980 10.111.156.10:59560      ESTABLISHED
tcp6       0      0 cs-892642682939-def:980 10.111.158.5:47392      TIME_WAIT
tcp6       0      0 cs-892642682939-def:980 10.111.158.8:49202      ESTABLISHED
tcp6       0      0 cs-892642682939-def:980 10.111.158.8:51194      ESTABLISHED
tcp6       0      0 localhost:970            localhost:55396          ESTABLISHED
tcp6       0      0 localhost:970            localhost:55410          ESTABLISHED
tcp6       0      0 cs-892642682939-def:980 10.111.156.10:60994      ESTABLISHED
tcp6       0      0 cs-892642682939-def:980 10.111.156.10:47658      ESTABLISHED
tcp6       0    494 cs-892642682939-def:980 10.111.158.8:50776      ESTABLISHED

Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags       Type       State      I-Node    Path
unix   3      [ ]       STREAM    CONNECTED  27394
unix   3      [ ]       STREAM    CONNECTED  25581
unix   2      [ ]       DGRAM     CONNECTED  20908
unix   2      [ ]       DGRAM     CONNECTED  25575
unix   3      [ ]       STREAM    CONNECTED  27306
unix   2      [ ]       DGRAM     CONNECTED  22237
unix   3      [ ]       STREAM    CONNECTED  27311
unix   2      [ ]       DGRAM     CONNECTED  30685
unix   3      [ ]       STREAM    CONNECTED  27959
unix   3      [ ]       STREAM    CONNECTED  21662    /var/run/docker/containerd/containerd.sock
unix   3      [ ]       STREAM    CONNECTED  26931
unix   3      [ ]       STREAM    CONNECTED  26667
unix   3      [ ]       STREAM    CONNECTED  27312
unix   3      [ ]       STREAM    CONNECTED  26672
unix   3      [ ]       STREAM    CONNECTED  27307
unix   3      [ ]       STREAM    CONNECTED  26666

```

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```

dgmidhun@cloudshell:~$ ss
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
u_str      ESTAB      0           0           * 27394                  * 27395
u_str      ESTAB      0           0           * 25581                  * 25580
u_dgr      ESTAB      0           0           * 20908                  * 20544
u_dgr      ESTAB      0           0           * 25575                  * 20544
u_str      ESTAB      0           0           * 27306                  * 27305
u_dgr      ESTAB      0           0           * 22237                  * 20544
u_str      ESTAB      0           0           * 27311                  * 27312
u_dgr      ESTAB      0           0           * 30685                  * 20544
u_str      ESTAB      0           0           * 27959                  * 27958
u_str      ESTAB      0           0           /var/run/docker/containerd/containerd.sock 21662 * 21661
u_str      ESTAB      0           0           * 26931                  * 26930
u_str      ESTAB      0           0           * 26667                  * 26668
u_str      ESTAB      0           0           * 27312                  * 27311
u_str      ESTAB      0           0           * 26672                  * 26671
u_str      ESTAB      0           0           * 27307                  * 27308
u_str      ESTAB      0           0           * 26666                  * 26665
u_str      ESTAB      0           0           * 26937                  * 26936
u_str      ESTAB      0           0           * 26353                  * 0
u_str      ESTAB      0           0           * 26686                  * 26685
u_str      ESTAB      0           0           * 26665                  * 26666
u_str      ESTAB      0           0           * 27961                  * 27960
u_str      ESTAB      0           0           * 20986                  * 20987
u_str      ESTAB      0           0           * 27308                  * 27307
u_str      ESTAB      0           0           * 26685                  * 26686
u_str      ESTAB      0           0           * 25580                  * 25581
u_str      ESTAB      0           0           * 27305                  * 27306
u_str      ESTAB      0           0           * 21655                  * 21656
u_str      ESTAB      0           0           /var/run/docker/containerd/containerd.sock 22113 * 22112
u_str      ESTAB      0           0           * 27957                  * 27956
u_str      ESTAB      0           0           * 26934                  * 26935
u_str      ESTAB      0           0           * 27399                  * 27398
u_str      ESTAB      0           0           * 26669                  * 26670
u_str      ESTAB      0           0           * 22112                  * 22113
u_dgr      ESTAB      0           0           * 19961                  * 20544
u_str      ESTAB      0           0           * 27396                  * 27397
u_str      ESTAB      0           0           * 24116                  * 24115
u_dgr      ESTAB      0           0           * 20813                  * 20544
u_str      ESTAB      0           0           * 26930                  * 26931
u_str      ESTAB      0           0           /var/run/docker/containerd/containerd.sock 21656 * 21655
u_dgr      ESTAB      0           0           /dev/log 20544 * 0
u_str      ESTAB      0           0           * 27310                  * 27309

```

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Non-authoritative answer:
*** Can't find ps: No answer

```

; <<>> DiG 9.18.39-0ubuntu0.24.04.3-Ubuntu <<>>
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 44521
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

```

```
## ANSWER SECTION:
259200 IN NS a.root-servers.net.
259200 IN NS b.root-servers.net.
259200 IN NS c.root-servers.net.
259200 IN NS d.root-servers.net.
259200 IN NS e.root-servers.net.
259200 IN NS f.root-servers.net.
259200 IN NS g.root-servers.net.
259200 IN NS h.root-servers.net.
259200 IN NS i.root-servers.net.
259200 IN NS j.root-servers.net.
259200 IN NS k.root-servers.net.
259200 IN NS l.root-servers.net.
259200 IN NS m.root-servers.net.
```

```
dgmidhun@cloudshell:~$
```

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```
;; WHEN: Wed May 20 06:33:53 UTC 2026
;; MSG SIZE rcvd: 239

dgmidhun@cloudshell:~$ pkill
pkill: no matching criteria specified
Try 'pkill --help' for more information.
dgmidhun@cloudshell:~$ kill
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
dgmidhun@cloudshell:~$ killall
Usage: killall [OPTION]... [--] NAME...
    killall -l, --list
    killall -V, --version

-e,--exact          require exact match for very long names
-I,--ignore-case    case insensitive process name match
-g,--process-group  kill process group instead of process
-y,--younger-than   kill processes younger than TIME
-o,--older-than     kill processes older than TIME
-i,--interactive    ask for confirmation before killing
-l,--list           list all known signal names
-q,--quiet          don't print complaints
-r,--regex          interpret NAME as an extended regular expression
-s,--signal SIGNAL  send this signal instead of SIGTERM
-u,--user USER      kill only process(es) running as USER
-v,--verbose        report if the signal was successfully sent
-V,--version        display version information
-w,--wait           wait for processes to die
-n,--ns PID         match processes that belong to the same namespaces
                    as PID
-Z,--context REGEXP kill only process(es) having context
                    (must precede other arguments)

dgmidhun@cloudshell:~$ bg
-bash: bg: current: no such job
dgmidhun@cloudshell:~$ fg
-bash: fg: current: no such job
dgmidhun@cloudshell:~$ jobs
dgmidhun@cloudshell:~$ nice
-10
dgmidhun@cloudshell:~$ renice
renice: not enough arguments
Try 'renice --help' for more information.
dgmidhun@cloudshell:~$
```

